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Abstract

Despite the large amount of data available concerning the Venice Lagoon, some morphological aspects have not been still investigated thoroughly. This is the case of the bathymetry of the 1930, edited by the Magistrato alle Acque, which was never used to make precise volumetric calculations. Only Rusconi (1987) compares the lagoon in 1930 and 1970, evaluating the difference in surface covering by bathymetric categories. The 1930 bathymetric data have been digitised and elaborated by GIS to make a 3D model to compare with the 1970 and 1990 digital products available for the lagoon. Using GIS is easy to perform precise volumetric calculations, thus obtaining information on the morphological evolution of the lagoon before and after the dredging of the Oil Channel.

The datum used in the 1930 map it's the National Altimetrical Net Zero (NANZ), established in 1910 by the IGM (Military Geographical Institute) using the observations of the marigraph at Campo S. Stefano. The NANZ corresponds to the average of the high and low tides recorded from 1884 to 1909, and assigned to the central year 1897 (Dorigo, 1961).

The hydrographic map edited in 1971 is referred to the datum level established in 1952 (still in use), corresponding to the mean sea level recorded at Genoa in the 1942 (IGM42).

In the works by Dorigo (1963) and Caputo *et al.* (1972), to uniform the data of the geodetic levelling of 1961 (referenced to the NANZ) with the new national altimetrical net (IGM42), 12.34 cm were subtracted from the elevation values of the 1961 survey. This value corresponds to the difference (measured in the 1961-62 geodetic levelling) between the elevations of the benchmark 48' (Ponte Crevada: line Conegliano-Venice) in the old and new geodetic net (Caputo *et al.*, 1972).

Considering the elevation difference of the benchmarks IGM 37 (church bell tower, Piazza di Spresiano, TV) and IGM 24' (Porta S.Tommaso, Treviso) surveyed in the 1908 (Dorigo, 1963) and in the 1968 (Caporali *et al.*, 1990), the obtained mean difference value would be 12.77 ± 0.37 cm.

To obtain the true morphological variation between the surveys, there is also to consider the drawdown caused by the subsidence between the 1897 (NANZ) and the first bathymetric survey (1927 - central year) and between this last and the 1970 (central year). Both these height differences must be summed to the difference between the reference levels.

To calculate the subsidence in the two time intervals between the three hydrographic surveys, an analysis has been carried out on the temporal series of the marigraph data of Venice and Trieste. The result of these analyses is a difference value of 10.1 cm. This value is confirmed by the elaboration made by Pirazzoli and Tomasin (2002) on the entire time series 1890-2000 of the Venice and Trieste data.

The differences obtained with different time trends are minimal, with a maximum range of 0.258 cm, and the mean value of 4 processing is the same as the one previously calculated. For this reason 10.1 cm is the value adopted as the subsidence correction.

Because the 1970 and 1990 bathymetric survey are referenced to the same datum, it is easier to apply the corrections to the 1930 survey:

a) lowering all the elevations by 10.1 cm (subsidence)
b) lowering all the elevations by 12.77 cm (different reference levels)
c) total correction = -22.87 cm = ~ 23 cm

Introduction

In the context of a wider study on the definition of the sedimentary budget, a complex work of rebuilding the historic hydrograph of the Venice lagoon has been carried out, digitising and elaborating in a vectorial format, in a GIS environment, the 1930 bathymetry, edited by the Magistrato alle Acque di Venezia in 1934 (Salvini, 1934). This gives, for the first time, a morphological dataset dating before the major environmental transformations (man-induced subsidence and the great channels digging), which have driven the biggest imbalances in the lagoon system. Subsequently the 1970 and 1990 bathymetries, supplied by the Consorzio Venezia Nuova, have been revised, in order to verify, through the comparison between the surveys, the reliability of the budget calculations found in literature and the quality of the 1990 survey.

Methodology

The 1930 bathymetric data have been digitised and elaborated using ESRI ArcGIS® software to make a 3D model to compare with the 1970 and 1990 digital products available for the lagoon. Using GIS is easy to perform precise volumetric calculations, thus obtaining information on the morphological evolution of the lagoon before and after the dredging of the Oil Channel.

The first step was the accurate vectorialization of the bathymetric maps edited by the Magistrato alle Acque in the 1930s. The 1930 maps consisted of a hundred of rasters (high resolution TIFs), courtesy of the Tide Forecast Centre of the Venice Commune. The maps have been georeferenced with ESRI ArcGIS® software and all the morphological objects, such as marshes, channels, islands and bathymetric points have been translated into vectorial objects, i.e. points, polylines and polygons. On the whole the created GIS objects are: a spot elevation shape, containing 78198 points; a channel edges shape, containing 2382 lines; a channel axes shape, containing 6039 lines; a land/water limit shape, containing 5468 lines.

Vectorial data from the 1970 maps (also edited by the Magistrato) were supplied by the Consorzio Venezia Nuova, which supplied their 1990 bathymetric data as well. All these data have then been used to build three 3D-models (TINs) of the Venice lagoon, using ESRI ArcGIS 3Danalyst® extension. Volume calculation has been carried out on these models; in order to quantify the morphological variations occurred during the two time intervals.

To obtain maps of the altimetrical variations the TINs have been converted into GRIDS of 50 m spacing cells and then two difference GRIDS have been calculated.

The Reference Datums

To perform the comparison between the two hydrographic maps edited in 1931 and 1971 (Rusconi, 1987) it's indispensable to fix a common reference datum, because each map is referred to the mean sea level of the National Altimetrical Net valid at the time of the survey.

Results

The evolution between 1930 and 1970

Owing to the previous considerations, all the vectorial elements of the 1930 lagoon model had been corrected (-23 cm), imposing a constant subsidence value for the whole lagoon. This is unlikely, because it's widely accepted that the phenomenon has had a different behaviour in different parts of the lagoon (Carbognin and Tosi, 2003). The simplification is compulsory though, because there are no available data on differential subsidence in the Venice Lagoon for the considered time period.

However this correction it's so hard on the marshes that the associated volume in 1930 (1,965,027 m^3) would be less then a third of those in 1970 (7,405,200 m^3). This is the only strong limit of the uniform correction, because it doesn't account for the marshes nature, which is capable of reacting to drawdowns trapping sediments to regain the equilibrium elevation (Favero, 1992).

To get around this error compensation volume has been introduced, for the marshes alone, equal to the difference between the two TINs, in the more than plausible event that in 1930 the marshes volumes were the same as in 1970. The difference between the marshes volume, in 1970 and 1930, is 5,440,173 m^3. Alternatively, it is possible to calculate the difference between the 1970 model and the original 1930 model, i.e. with no correction, following the hypothesis that the marshes elevations remained the same despite the subsidence. In this latter case a further correction is needed, of 1,080,645 m^3.

On the basis of the conducted analysis the sedimentary budget, 1970-1930, amounts to about -83.5 millions of m^3, something more then -90 millions of m^3 considering the marshes compensation volumes. Subtracting from this latter value the excavation volume of the industrial area and of the Oil Channel (42 millions of the m^3), the erosion deficit amounts to about 48 millions of m^3 in 43 years (1927-1970), which corresponds to an erosion rate of 1,117,000 m^3/year.

Fig. 1 represents the result of the comparison between the two bathymetric survey obtained from the processing of the GRIDs (with the 1930 corrected for 23 cm), through subtraction of the mean depth values in 50 m spacing cells. In the comparison the channels affected by the large digging works performed from 1952 to 1969 are clearly visible.

Using the produced GRIDs is also possible to calculate the area and volume variations occurred in specific bathymetric classes (Table 1).

The greater surface reductions regarded the marshes (> 0 m), with almost 15 millions of m^2 lost in 43 years, the uppermost intertidal zone from 0 to -0.25 m (-11 millions of m^2) and the upper subtidal zone from -0.50 to -0.75 m (-23.7 millions of m^2).

Table 1: Area and volume balances of the whole lagoon, divided by altimetrical categories (1970 – 1930)

	Area balance (1970-1930)			**Volume balance**
Class (m)	**Area (m^2 *10^6)**	**Δd**	**%**	**Δ Volumes**
marshes (> 0.25)	-5.647	-0.036	-49.00%	-1,973,270
0 ÷ 0.25	-9.385	0.096	-20.39%	-22,265
-0.25 ÷ 0	-11.362	-0.036	-14.16%	-323,149
-0.5 ÷ -0.25	3.490	0.003	4.12%	-1,236,390
-0.75 ÷ -0.50	-23.595	0.012	-30.70%	16,039,228
-1.00 ÷ -0.75	13.285	-0.011	36.28%	-12,371,677
-1.25 ÷ -1.00	16.870	0.000	113.13%	-18,745,648
-1.50 ÷ -1.25	6.870	0.006	77.13%	-9,220,060
-2.00 ÷ -1.50	2.987	-0.008	32.27%	-5,253,627
-5 ÷ -2	1.865	0.074	8.93%	-4,036,751
-10 ÷ -5	1.440	-0.111	9.59%	-11,946,639
-15 ÷ -10	1.965	-0.515	47.44%	-25,665,848
-20 ÷ -15	0.982	0.504	185.38%	-15.993.516
-25 ÷ -20	0.0125	0.117	8.06%	-253,557
< -25	-0.0275	0.550	-22.45%	947,775
Total volume difference (1970-1930), in m^3 →				-90,055,396
Total eroded volume (net of the industrial canalization= 42.000.000 m^3), in m^3 →				-48,055,396
Erosion rate, in m^3/year (on 43 years: 1970-1927) →				**-1,117,567**

All the other bathymetric categories, except the deepest channels (< 25 m), feature a surface increment which, thus testifying the general deepening of the lagoon. It must be noted that the area increase of the channels category, though apparently modest, is responsible for more than 50% of the total volume budget. Nevertheless this percentage must be ascribed, almost totally, to the excavations in the industrial area and along the Malamocco-Marghera channel.

The evolution between 1970 and 1990

The calculated difference is 19.8 millions of m^3, corresponding to an erosion rate of about 990,000 m^3/year.

Yet there is a discrepancy in the 1990 marshes mean elevation. In the original data the area with a mean elevation greater than 25 cm seems to have been tripled in 20 years, with an increase in the mean elevation of 15 cm. The data is certainly incorrect and must be probably ascribed to the survey method used in 1990 (aerophotogrammetry) for the emerged morphologies. Assuming that the emerged surface extent is correct, it is logical to consider the mean elevation as unchanged compared with the 1930 and 1970 surveys (0.16 cm). Consequently, assigning the indicated mean elevation to the whole subaerial morphologies, the erosion budget increases to 24.4 millions of m^3 in 20 years, corresponding to a rate of about 1,222,000 m^3/year (Table 2).

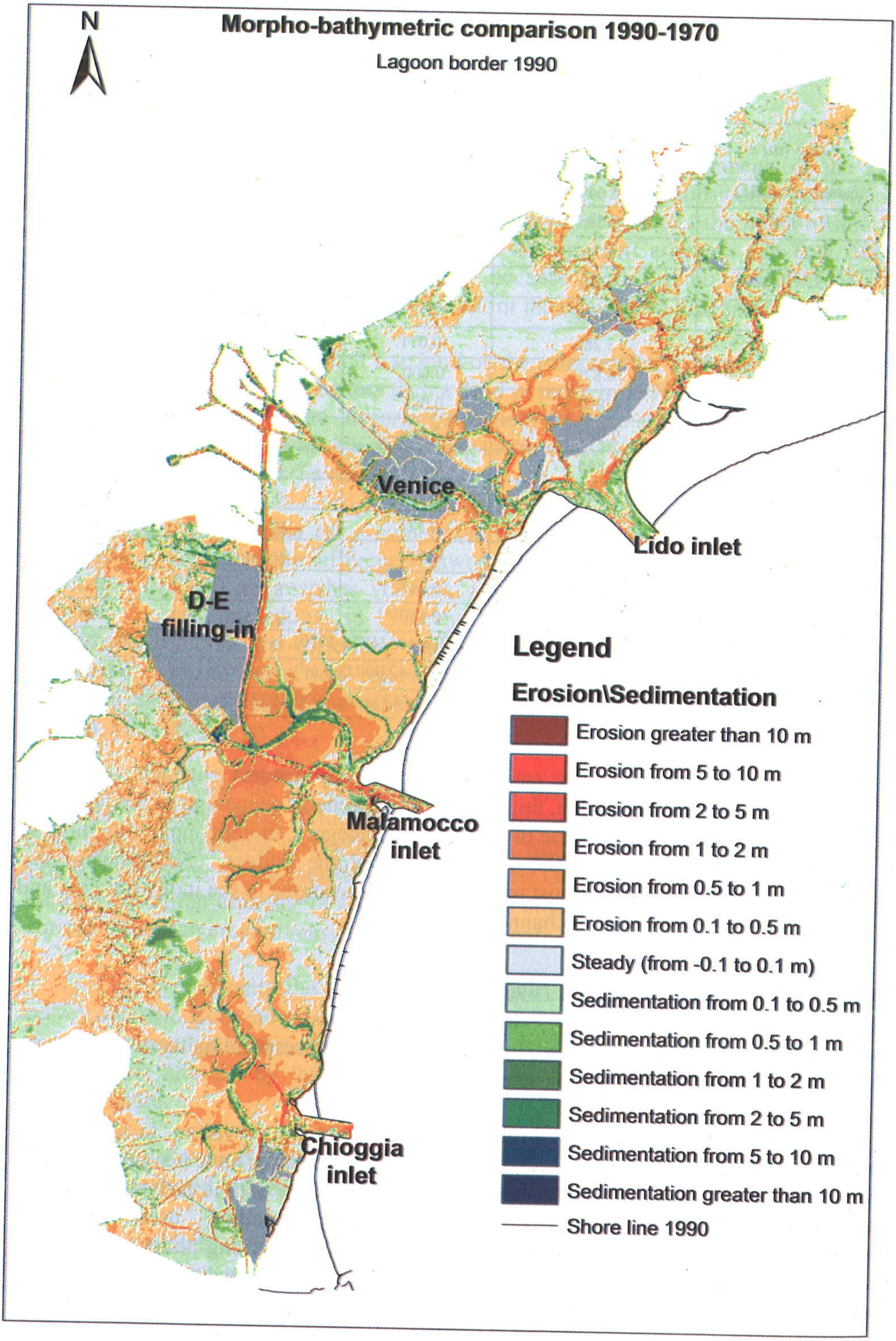

Fig. 1: Altimetrical variation map of the Venice Lagoon (1930-1970).

Table 2: Area and volume balances of the whole lagoon, divided by altimetrical categories (1990 – 1970)

	Area balance (1990-1970)			Volume balance	
Class (m)	**Area (m^2 *10^6)**	**Δd**	**%**	**ΔVolumes**	**ΔV corr.**
Marshes (> 0.25)	(10.882)	0.151	-13.47%	(5,987,522)	**-604,710**
From 0 to 0.25	(-16.610)	-0.002		(-2,131,461)	
From -0.25 to 0	-1.422	0.004	-4.01%		293,188
From -0.5 to -0.25	1.342	0.003	2.96%		-353,189
From -0.75 to -0.50	-9.645	0.005	-14.47%		6,401,298
From -1.00 to -0.75	-17.825	0.000	-21.90%		15,570,313
From -1.25 to -1.00	-4.787	-0.009	-9.36%		4,918,619
From -1.50 to -1.25	15.300	-0.017	74.48%		-21,311,839
From -2.00 to -1.50	19.865	0.008	118.88%		-33,880,462
From -5 to -2	4.930	0.082	19.75%		-12,764,357
From -10 to -5	-0.995	-0.078	-5.86%		5,822,851
From -15 to -10	-1.047	0.030	-16.08%		12,595,897
From -20 to -15	-0.062	-0.222	-4.06%		707,874
From -25 to -20	0.057	-0.078	33.82%		-1,268,635
< -25	0.017	-0.104	17.50%		-570,804
Total volume differences (1990-1970), in m^3 →				-19,983,185	-24,443,956
Erosion rate, in m^3/year (on 20 years: 1990-1970) →				**-999,159**	**-1,222,198**

The actual validity of these calculations is tied to the correctness of the compared surveys. While it was possible to check the reliability of the 1970 survey, through a comparison with the corresponding CTR (Regional Technical Map), there are some doubts about the 1990 data. For example in Fig. 2 is represented a zoomed zone of the 1990 TIN in which geometrical artefacts are visible, caused by poor data density. These artefacts compromise the volume calculations and the gridding, so that the resulting numbers are not so reliable as they could be.

In Fig. 3 the result of the comparison between the 1990 and 1970 GRIDs is reported. The greatest changes have taken place in the Malamocco basin. It is easily noted the lowering occurred in the subtidal zones adjacent to the reclaimed areas D-E, near the intersection between the Oil Channel and the old feeder channel of the Malamocco inlet. The deepenings are comprised between 1 and 2 m, and represent the peak of a phenomenon that involves the whole lagoon basin, i.e. the deepening of the subtidal zones, with values comprised between 0.1 and 2 meters.

The extension of the marshes is reduced and presents a greater fragmentation than in 1970. These areas are characterized by deepenings lower than a meter, and are moreover surrounded by sedimentation zones (within 0.5 m).

The channel areas are the zones with the greatest sedimentations (within 5 m): mainly the old feeder channel of Malamocco inlet, the Giudecca channel and the feeder channel of Chioggia inlet.

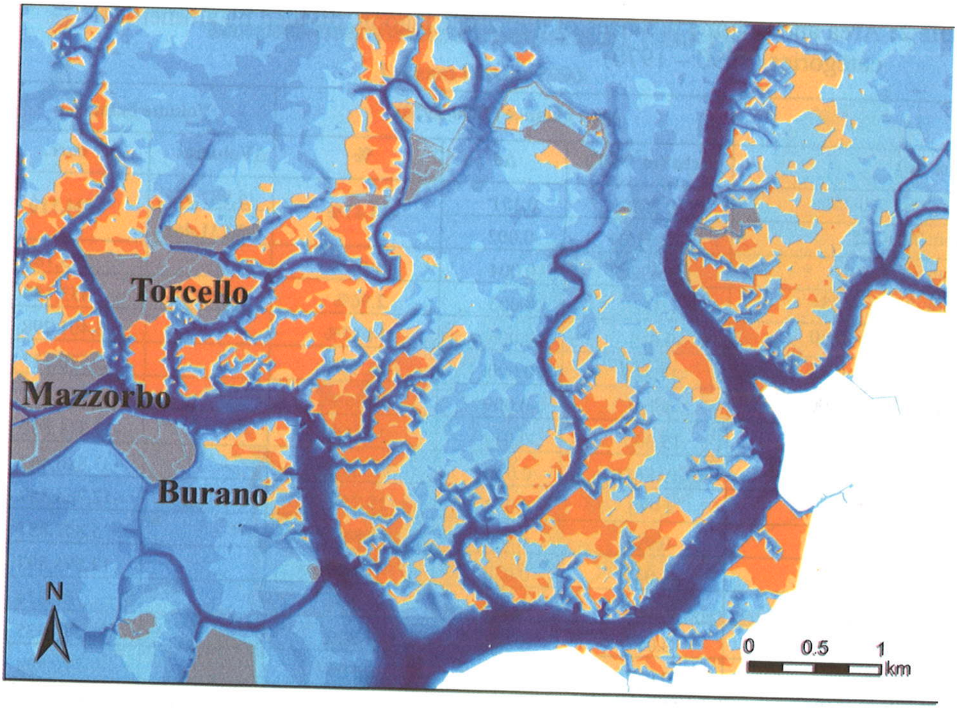

Fig. 2: Zoomed area of the 1990 TIN, with visible geometric artefacts

Conclusions

Even with all the aforementioned cautions, the results obtained from the volume comparisons, in the 1930-1990 period, show believable evolution trends, well pictured in the two altimetrical transformation maps.

The different behaviour through time of the lagoon basin can be well represented by Fig. 4. Besides the percentage distribution of the areas covered by the different bathymetric categories in the three surveys examined[1], the figure reports the trend of the area variations occurred in the 1930-70 and 1970-90 time intervals.

During the period 1930-70 the greatest area reductions involved the marshes and the altimetrical categories within -0.75 m. The surface expansion of the other categories (< -0.75 m) can be related to the increase of depth, as for the subtidal zones, while, for the channels, to the effects of the large excavations carried out in the industrial area.

The fulcrum of the system variations, represented by the double arrow (Fig. 4), is the -0.75 m depth. Therefore it is possible to say that between the 1930s and the 1970s the lagoon suffered a general, almost undifferentiated, "drowning".

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[1] The 1930 is to be reported corrected by the subsidence, except for the marshes.

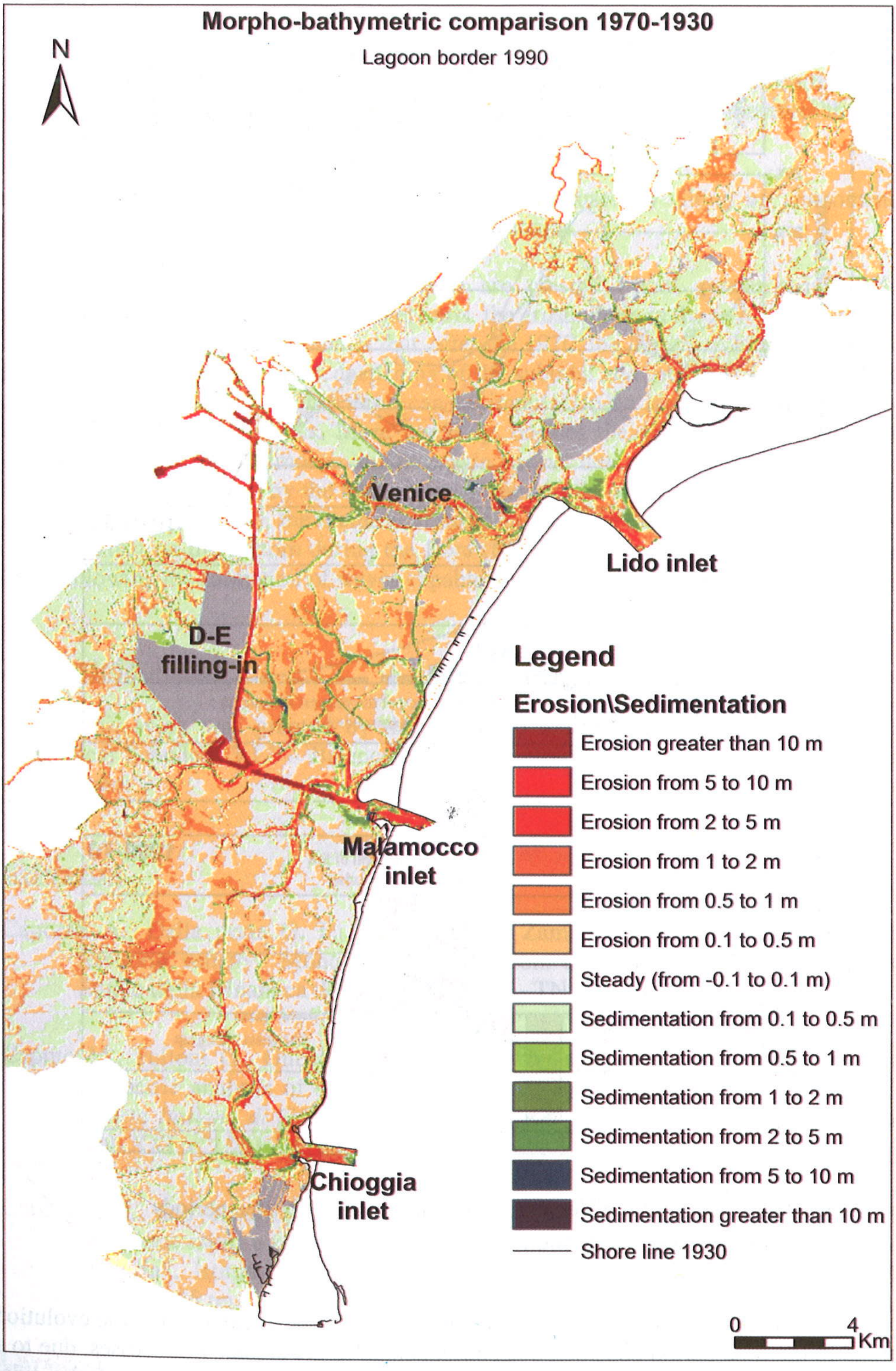

Fig. 3: Altimetrical variation map of the Venice Lagoon (1970-1990)

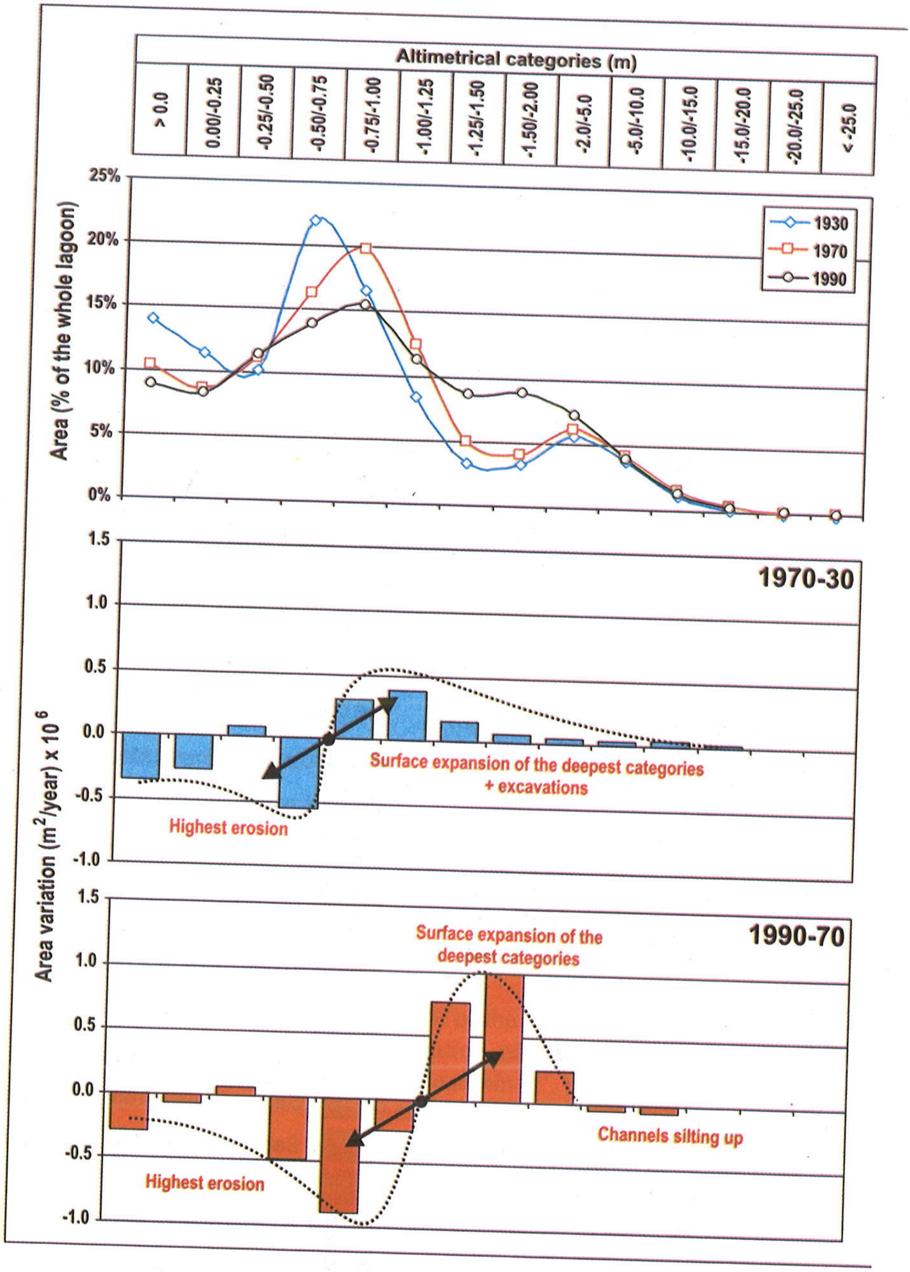

Fig. 4: Evolution of the Venice Lagoon during the two time periods (1930-1970, 1970-1990)

Although the subsidence has been eliminated from the calculation, the evolution trend in the time period considered is consequential to the erosional processes, due to a transgressive effect, which have determined an effectiveness increase of the tidal currents and an overall increase in tidal prism.

The following period (1970-90) witnesses a spreading of the erosional processes which leads to an enlargement of the surfaces covered by the deeper altimetrical categories, within the -5 m limit. The most involved zones in this phenomenon are the subtidal ones comprised between -0.5 and 1.25 m. This latter depth is therefore the new fulcrum of the lagoon system.

The surface reductions of the channels are caused by silting up phenomena, which involve above all the central and southern basins.

Finally the surface transformations are on the whole greater than in the preceding period, although, on a budget basis, the erosion rate in the two periods is very similar. This allows the hypothesis that, after the 1970, the lagoon sedimentary regime has changed, with a significant deepening of the subtidal and intertidal zones (between 50 and 125 cm of depth). This trend can be imputed to an increase of the erosional effects due to hydrodynamics. More precisely, besides the amplification of the tidal currents effect in the Malmocco and Chioggia basins, due to the new canalisations, a significant role is played by the currents associated to the wave motion. As previously stated by Cavaleri (1980), the increase of depth in the more extensive areas reduces the friction effect on the bottom during the Bora (NE wind) events, i.e. those with greater fetch, allowing the formation of waves capable of triggering erosive bottom flows.

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